Three Routes to Failure: An Interpretable Diagnosis of Task Difficulty and Negative-Set Contamination in Genomic Sequence-Classification Benchmarks

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***Abstract*—Leaderboards of genomic foundation models (FMs) rarely ask what makes a sequence-classification task hard or whether a score reflects biology or a dataset-construction artifact. We provide a CPU-only diagnostic framework that route-labels benchmark tasks from interpretable k-mer spectra and gradient-boosted trees; we develop it on two suites, apply it with fixed thresholds to the 28 datasets of a third suite, GUE (where only its positional labels survive post hoc checks), and release composition-equalized enhancer splits and a diagnostic tool. It outputs three routes to failure: *shared-motif* (abundant motifs shared by both classes), *positional* (a fixed-position motif a spectrum discards), and *contamination* (a compositionally distinct negative set), each established by intervention. Motif presence does not predict success; in a semi-synthetic dose-response, discriminability does, causally. The negatives of two separately curated human enhancer benchmarks carry 20.9–22.7 percentage points more TATA-box sequences than their positives; coordinates refute core-promoter leakage in one set (underpowered in the other) and trace the gap mainly to GC-rich positives, with genomic negatives drawn from gene-distal or unmarked DNA. Against dinucleotide-shuffled negatives the negative-class TATA skew disappears. Two FMs (HyenaDNA, DNABERT-2), frozen or fine-tuned, lose significant enhancer AUROC when composition-biased negatives are removed, not when random ones are; a GC-balanced control (HyenaDNA) also loses some. The contamination is a benchmark-design confound, not dataset corruption.**

***Index Terms*—genomic benchmarks, interpretable machine learning, negative-set contamination, DNA foundation models, k-mer features**

I. Introduction

Benchmark leaderboards now drive much of the development of DNA foundation models (FMs) [1]–[5]: a model's standing is its aggregate accuracy across suites such as Genomic Benchmarks (GB) [6] and the Nucleotide Transformer (NT) downstream tasks [1], yet the benchmarks themselves are treated largely as black boxes. We rarely ask *what property of a task makes it easy or hard*, or whether a high score reflects biological signal rather than an artifact of dataset construction.

We provide a CPU-only diagnostic framework that route-labels genomic sequence-classification benchmarks, developed on GB and the NT downstream tasks and applied with thresholds fixed in advance to the 28 datasets of the Genome Understanding Evaluation (GUE) [3], most of which share sequences with GB/NT. It treats a simple classifier as an **interpretable lens**, not a competitor to FMs: k-mer spectra classified with gradient-boosted trees (LightGBM [7]), so that every prediction traces back to short motifs. Pretrained FMs, frozen and end-to-end fine-tuned, enter only as probes of whether models exploit a dataset shortcut and are never claimed to be beaten. The output is a **mechanistic taxonomy** of *why* a task fails: **shared-motif** (Route 1; abundant motifs occur in *both* classes), **positional** (Route 2; a fixed-position motif that a position-blind spectrum discards), or **contamination** (Route 3; a *negative* class drawn from a different compositional distribution, here AT-rich relative to the GC-rich enhancer positives).

Contributions. (1) A diagnostic that route-labels the tasks of two benchmark suites (Table I); applied unchanged to GUE, it labels splice and, against our pre-stated prediction, both TATA-promoter tasks positional. (2) Each route established by intervention: Route 1 by a dose-response at arbitrary sample size, Route 2 by a selective, bin-graded rescue, and Route 3's composition-associated magnitude by dinucleotide-shuffled negatives and by composition-equalized cleaning benchmarked against size-matched random negatives, with a control-motif panel showing that the negatives' TATA skew is composition-driven, not a scan artifact. (3) Two architecturally unrelated FMs lose significant enhancer AUROC when composition-biased negatives are removed, as frozen probes and after end-to-end fine-tuning, while removing as many at random costs them nothing (on a GC-balanced control HyenaDNA also loses some, except as a frozen summary-token probe). (4) A robustness analysis (Table II), composition-equalized enhancer splits, the frozen-FM embedding cache and a diagnostic command-line interface (CLI).

Throughout, *contamination* labels a failing task that passes the *TATA gate* (Section III; chiefly, $\geq$10 percentage points more TATA boxes in the negatives than in the positives). This compositional shift supplies much of the task's separability; the label names a benchmark-design confound, not a claim of core-promoter leakage (which our coordinates test).

II. Related Work

DNA foundation models, benchmark suites and baselines. Genomic FMs — DNABERT [2], DNABERT-2 [3], the Nucleotide Transformer [1], HyenaDNA [4], Caduceus [5] —

are pre-trained on large genomic corpora and ranked on classification suites: Genomic Benchmarks [6], the NT downstream tasks [1] and GUE [3]. From-scratch convolutional networks are competitive with genomic FMs [8], and embedding-plus-classifier baselines [9], [10] question how much pretraining adds; we analyze the *benchmarks* instead, using a simple, interpretable model to *explain why* tasks are easy or hard.

Negative-set composition bias. The k-mer SVM of Lee *et al.* [11] and the gapped-k-mer SVM [12] select features and controls to avoid trivial composition cues, and their tooling generates GC-, length- and repeat-matched negatives [13]. DeePromoter [14] builds harder negatives because TATA-like motifs are more common genome-wide than in true promoters; Krützfeldt *et al.* [15] compare GC-matched genomic negatives with k-mer-preserving shuffles (both among our interventions) and find that the negative set determines what a model learns. Rather than choosing negatives to train a better predictor, we diagnose composition contamination *on the original benchmark splits*, with coordinates, interventions and FM probes.

III. Methods

The diagnostic pipeline is CPU-only, uses a fixed random seed (42) unless stated, and reports 95% percentile-bootstrap confidence intervals (CIs; 1000 resamples, the test sequence as the resampling unit unless stated; cross-task statistics resample tasks). Software versions, hyperparameters and the pinned dataset revision are in `REPRODUCIBILITY.md` and per-analysis READMEs; the Supplementary Material (SM) [16] gives the full analyses (SM S1–S9).

Featurization and models. Each sequence is its k-mer spectrum ($k \in \{3,4,5,6\}$; counts of all 4^k ACGT k-mers, L1-normalized). Per dataset we select k on a stratified 80/20 validation carve of the training set, retrain on the full training split, and evaluate once on the test split. The main model is LightGBM [7] with hyperparameters fixed across datasets, with a scikit-learn [17] logistic-regression (LR) floor; a *composition-only baseline* is LightGBM on the 4 mono- and 16 dinucleotide frequencies alone. k-mer spectra are the established interpretable representation for regulatory sequence [11], [12]; boosted trees capture non-linear k-mer effects on a CPU. Robustness refits use L1/L2 LR tuned by validation MCC (MCC-tuned LR) and an explainable boosting machine (EBM) [18] on composition plus 128 train-selected k-mers. We report the Matthews correlation coefficient (MCC) [19], accuracy and the area under the ROC curve (AUROC).

Motif enrichment and discriminability. We score k-mers against JASPAR 2024 CORE [20], [21] log-odds position-specific scoring matrices (PSSMs) and test, per transcription factor (TF), whether its motif score is enriched among high-gain k-mers (gain-weighted statistic, analytic permutation null, Benjamini–Hochberg false discovery rate (FDR) [22] $q < 0.05$). The *motif-only AUROC* is the AUROC of the both-strand hit count above a fixed bits-per-position threshold (GC-box 1.0; TATA 1.3, since TBP is a low-information AT-rich motif; conclusions hold over 0.8–1.5; one vertebrate TBP profile for all species), and a task's *discriminability* is its highest TF motif-only AUROC; TATA-box fractions use a seeded sample of ≤4,000 sequences per class.

Dose-response and position-binned spectra. Into the positives only of random ACGT backgrounds matched in length and composition to a real task, we implant instances sampled from a JASPAR TF's position weight matrix (PWM; SP1, MYC, TBP) at a tunable rate. This dials discriminability, the realized motif-only AUROC d (same scan), and is silent on motif presence; we record test MCC (five seeds). The *position-binned spectrum* (Route 2) splits each sequence into $B \in \{2,4,8,16\}$ equal bins whose L1-normalized spectra are concatenated to the global spectrum (B=1 is the baseline).

Contamination cleaning (Route 3). Only negatives are cleaned, in `human_enhancers_cohn` (`cohn`), in `nt_enhancers` and in the *GC-balanced control* `drosophila_enhancers_stark` (`drosophila`; failing, TATA skewed to the positives). (i) *TATA-flag*: remove a seeded subset of TATA-flagged negatives (TBP ≥1.3 bits/pos) equal to the class excess; (ii) *GC-matched*: subsample negatives to the positive GC histogram; (iii) *composition-equalized* (*matched*): subsample negatives within each split to the positives' histogram of an LR propensity score on the same 20 frequencies [23]. We report k-mer AUROC against size-matched random negatives (10 seeds) along a strictness sweep (α: share of the propensity-histogram mismatch removed) and a caliper sweep (two-sided 1:1), with the model or the original test held fixed, and in overlap-weighted [24] and positive-side designs (SM S4). *Dinucleotide-shuffled negatives* are Altschul–Erickson shuffles [25] of the positives (one random draw; `cohn` TATA AUROC 0.549–0.554 across 16 draws). The *control-motif panel* scans with 20 column-permutations of the TBP PSSM and 10 random PWMs matched to its information content.

Route labels. Fixed rules, in order: *solved* at MCC ≥0.7; else *positional* if B=8 raises MCC significantly by >0.2; else *contamination* if the TATA gate fires (negative-class TATA-box excess ≥10 points, TATA motif-only AUROC <0.5, moved toward 0.5 by TATA-flag cleaning); else *shared-motif* if motifs are abundant (>0.2 hits per sequence per TF, on average) yet no TF motif-only AUROC reaches 0.6; else no single route. Exceptions: the three GB `demo`/`dummy` tasks are *composition* by task type, and multiclass `nt_enhancers_types` (no single route by the rules) is *shared-motif* by hand.

Coordinates and provenance (Route 3). We recover GRCh38 coordinates for the negatives (exact substring for `cohn`; minimap2 [26] for `nt_enhancers`), intersect them with GENCODE v44 transcription start sites (TSS, ±500 bp; Fisher test, TATA-to-TSS offset), and compare the negatives with random and rule-drawn genomic backgrounds (GENCODE genes, ChromHMM [27], isochores, ENCODE candidate cis-regulatory elements (cCREs) [28]; SM S3).

Homology control. We re-cluster train+test of all 15 tasks with MMseqs2 [29] (both strands; ≥80% identity over ≥80% of both sequences) and compare purging test sequences with a train homolog against class-matched random removal, and cluster-aware with random re-splits (three split seeds; SM S6).

FM probes. *Frozen:* embeddings of HyenaDNA-tiny and

DNABERT-2-117M from CPU forward passes (mean pooling and the summary token, [SEP] for HyenaDNA and [CLS] for DNABERT-2), with LR and LightGBM heads; the embeddings are released. *Fine-tuned:* HyenaDNA-tiny (all 0.44M parameters) and DNABERT-2-117M (LoRA [30], 1.25% trainable), a linear head on mean-pooled states, one fixed recipe each, seeds 42–44 (the only GPU step). Each scores the original test positives against the original, composition-equalized and size-matched random test negatives, and is retrained on the latter two; contrasts were fixed in advance; the shares of above-chance AUROC lost are post hoc (SM S5).

IV. Results

A. Benchmark landscape

Achievable MCC (Table I) spans 0.244 to 0.914, and "solved" tasks are separated from "failed" ones by non-overlapping CIs. The same gradient appears in both suites: regulatory and promoter tasks are well-solved; enhancer, open-chromatin and splice-site tasks are not. These MCCs are stable across seeds (five seeds; max SD 0.013 over eight key tasks).

B. Route 1 — discriminability, not presence

We hypothesized that well-solved tasks are those whose high-gain k-mers match many known TF motifs. **Disconfirmed**: the number of FDR-enriched TF motifs does not track learnability (Spearman $\rho = -0.03$ over the nine GB tasks, n.s.). The *failed* enhancer/open-chromatin tasks are in fact the *most* motif-rich (`human_enhancers_ensembl` 265, `human_ocr_ensembl` 243 FDR-enriched TF motifs). Motif **discriminability** is what the causal dose-response below ties to learnability. In `human_nontata_promoters` (whose solved MCC depends partly on train/test homologs; Table I†), GC-box/Sp1–KLF motifs concentrate in the positive class (SP1 ranked first; SP2 motif-only AUROC 0.751), whereas on the failed `human_ocr_ensembl` the dominant E-box/AP-1 motifs occur in *both* classes (TFAP4 AUROC 0.527, BATF3 0.573) — abundant but non-discriminative.

Causal dose-response (Fig. 1). Across $n = 135$ settings (3 motifs × 5 seeds × 9 doses), MCC rises monotonically with d (Spearman +0.924). Because these clustered settings are not independent, the load-bearing statistic is the agreement of *three independent per-motif slope fits*: +1.97 (SP1, 95% CI $[1.87, 2.07]$), +2.02 (MYC, $[1.90, 2.14]$), +1.96 (TBP, $[1.78, 2.13]$) — all excluding 0 and overlapping, so the dose-response is a property of task structure, not one motif's identity (pooled least-squares slope +1.99, $[1.92, 2.06]$). Only this synthetic result isolates discriminability.

Real-task consistency. On the 11 TF-motif tasks (all but splice and the three composition tasks) discriminability tracks learnability (Spearman $\rho = +0.74$, exact $p = 0.013$; leave-one-task-out (LOTO) mean absolute error (MAE) 0.14 MCC, R^2 +0.44 $[-0.27, +0.70]$), but, post hoc, GC content tracks it at least as closely ($\rho = +0.86$, exact $p = 0.001$; ρ difference $[-0.32, +0.53]$; the two are collinear) and alone separates, narrowly, the solved promoter/regulatory from the failing enhancer/open-chromatin tasks. In a pre-specified multivariate analysis (nine covariates, including length, within-class GC SD and train/test homology) no covariate set predicts better than discriminability alone (the selected GC-mean models tie); test noise explains only 1.5% of the LOTO misfit, the rest being between-task differences that, post hoc, follow the promoter/enhancer split. The tasks are 9 distinct sequence sets once exact copies and unions are collapsed; the correlation is not significant with LR or an EBM, nor on the pre-specified cluster-aware homology re-split (+0.55, n.s.; random re-split +0.57; Table II; SM S2). The real-task relation is consistent with the dose-response, not independent evidence for it.

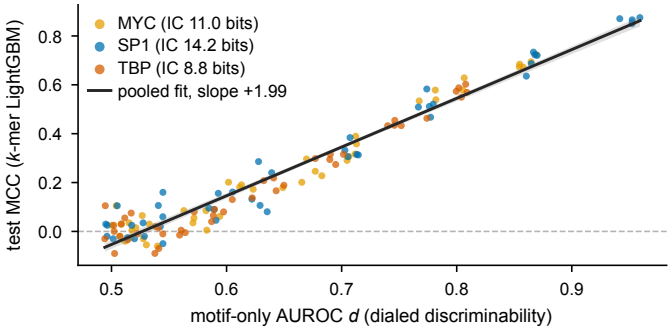


Fig. 1. Route 1 dose-response (semi-synthetic): test MCC vs. the dialed motif-only AUROC d of implanted TF motifs ($n = 135$: 3 motifs × 5 seeds × 9 doses; IC: information content), with the pooled least-squares fit and its 95% bootstrap band; the per-motif slopes overlap.

C. Route 2 — positional signal, selective and graded

Splice-site classification fails (`nt_splice_sites_all` MCC 0.244; motif-only AUROC 0.522, at chance), consistent with a fixed-position GT–AG consensus the position-blind spectrum discards. Adding a position-binned spectrum ($B = 8$) gives a clean **selective rescue** (paired bootstrap): splice recovers almost entirely, MCC 0.244→0.868 ($\Delta = +0.624$, 95% CI $[0.594, 0.651]$). Across all 15 GB/NT tasks the pattern is *selective*: one large rescue, and $|\Delta| \leq 0.028$ for every other task, including the fixed-position `nt_promoter_tata` (+0.019, n.s.). Within the responding task the effect is *graded* in bin resolution ($B \in \{2, 4, 8, 16\}$: +0.46, +0.53, +0.60, +0.66 on a 20,001-sequence training subsample), not one lucky bin. On GUE the positional label goes to splice (ΔMCC +0.41) and, against our pre-stated prediction, to both TATA-promoter tasks (+0.23, +0.25); one shares its positives with `nt_promoter_tata`, which gains only +0.02, so the route follows how the negatives were built (SM S8).

D. Route 3 — negative-set contamination

Existence and replication. The strongest single-motif separator on the failed enhancer datasets is TATA/TBP, skewed to the **negative** class. On `human_enhancers_cohn`, negatives carry far more TATA boxes than positives (motif-only AUROC 0.371, 95% CI $[0.365, 0.376]$; TATA-box fraction positive 34.6% vs. negative 57.3%, an excess of 22.7 percentage points). The same signature appears on the separately curated `nt_enhancers` (AUROC 0.393 $[0.387, 0.399]$; TATA-

TABLE I

BENCHMARK LANDSCAPE: BEST-k LIGHTGBM PER TASK, BY MCC (95% BOOTSTRAP CIS ON THE TEST SPLIT; AUROC ONE-VS-REST MACRO FOR MULTICLASS TASKS).

Suite	Task (dataset id)	n_{train}	n_{test}	k	MCC [95% CI]	AUROC	Route (Section III)
GB	demo_human_or_worm	75,000	25,000	5	0.914 [0.909, 0.919]	0.992	composition
GB	human_ensembl_regulatory	231,348	57,713	5	0.907 [0.904, 0.910]	0.991	solved
NT	nt_promoter_all	53,276	5,920	4	0.888 [0.877, 0.899]	0.985	solved
NT	nt_promoter_no_tata	47,767	5,299	4	0.887 [0.873, 0.899]	0.985	solved
GB	human_nontata_promoters[†]	27,097	9,034	6	0.874 [0.864, 0.883]	0.989	solved
NT	nt_promoter_tata	5,509	621	4	0.846 [0.803, 0.886]	0.984	solved
GB	demo_coding_vs_intergenomic_seqs	75,000	25,000	5	0.800 [0.793, 0.808]	0.965	composition
GB	human_enhancers_ensembl[a]	123,872	30,970	6	0.593 [0.585, 0.602]	0.875	contam. (weak)
GB	dummy_mouse_enhancers_ensembl	968	242	6	0.529 [0.422, 0.632]	0.858	composition
NT	nt_enhancers[b]	14,968	400	4	0.514 [0.432, 0.595]	0.850	contamination
NT	nt_enhancers_types[c]	14,968	400	6	0.476 [0.415, 0.532]	0.837	shared-motif
GB	human_enhancers_cohn	20,843	6,948	5	0.461 [0.440, 0.482]	0.805	contamination
GB	human_ocr_ensembl	139,804	34,952	6	0.421 [0.412, 0.430]	0.785	shared-motif
GB	drosophila_enhancers_stark	5,184	1,730	4	0.398 [0.352, 0.441]	0.763	shared-motif
NT	nt_splice_sites_all	27,000	3,000	5	0.244 [0.219, 0.271]	0.704	positional

[†]Relabels under cluster-aware re-splits. [a]Weak: narrow TATA-gate pass (10.8 points); not coordinate-mapped or FM-probed; 76% of test positives are exact training copies (purged MCC 0.43, unadjusted). [b]100 of its 400 test sequences, all of them positives, are such copies (leak-free AUROC 0.735). [c]Its class-2 (strong-enhancer) test sequences are these same 100 copies.

box fraction 10.6% vs. 31.5%, excess 20.9 points) — so the contamination is **not GB-specific**. The negatives are systematically more AT-rich (GC-content AUROC 0.737/0.822 for cohn/nt_enhancers; Fig. 2).

Provenance of the negatives. cohn negatives are random 500-bp windows ≥15 kb from genes and outside liver H3K27ac peaks [31] (94.7% lie ≥15 kb from any protein-coding gene vs. 46.2% of random windows, though 32% lie in introns, mostly of non-coding genes). Of the nt_enhancers training negatives 78.7% are synthetic, generated by RankGAN [32] (as are 68.8% of its training positives); the genomic ones (iEnhancer-2L [33]) are ChromHMM low-signal state 13 in all nine ENCODE cell lines [27] (99.8% vs. 37.0% of random windows), whereas the test negatives (iEnhancer-EL [34]), whose FASTA headers point to single state-13 segments rather than the nine-line intersection, are state 13 in all nine lines in only 64.0%. Most of both genomic negative sets lie in AT-rich isochores [35] (71%/84% vs. 60%), yet sampling explains only 20%/31% of the GC gap (cohn/genomic nt_enhancers; 13.5% over the whole, mostly synthetic class): the rest is the positives' GC-richness.

Mechanism: a composition gap, not TSS leakage. Recovering GRCh38 coordinates for all 13,896 cohn negatives **refutes** our core-promoter-leakage prior (criteria fixed in advance): TATA-flagged negatives overlap a TSS (±500 bp) in only 1.8% of cases, *below* the 3.3% of non-flagged negatives (0.55× depletion, Fisher p=3×10^{-8}; positives 15.6%); flagged negatives near a TSS show no peak at the canonical −25 to −30 bp (median −96). The excess is thus compositional, mostly the positives' GC-richness, while the sampling rules shift the negatives toward AT relative to random genome (TATA +3.3/+7.0 points; backgrounds re-drawn by each dataset's stated rule, with unstated details inferred post hoc, reproduce their GC). Only 28.2% of nt_enhancers sequences are genomic (the verbatim iEnhancer originals, including the test split); the rest are RankGAN-synthetic, with

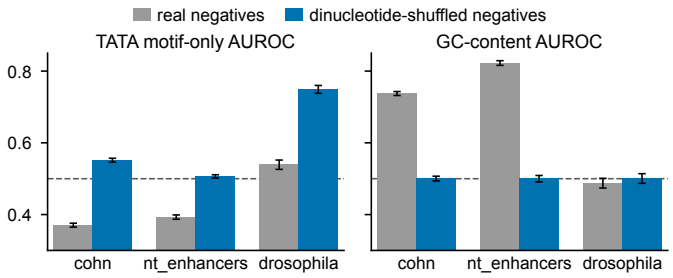


Fig. 2. Route 3: TATA motif-only (left) and GC-content (right) AUROC against real negatives (grey) and dinucleotide-shuffled negatives (blue; shuffles of the positives), with 95% bootstrap CIs. For the contaminated sets the negative-class TATA skew *vanishes* once the composition gap is removed; for the GC-balanced control (drosophila), whose TATA is a genuine positive-class motif, the TATA AUROC instead rises.

no locus. On the genomic subset the flagged class is likewise TSS-*depleted* (0.6% vs. 1.4%, 0.46×), with no offset peak: directionally corroborative but underpowered (p=0.17).

Magnitude: shortcut removal or sample loss? Composition equalization matches the negatives' GC + dinucleotide distribution to the positives' (max. standardized mean difference 1.007→0.059 for cohn; the one-sided equalizer, which only subsamples negatives, cannot balance nt_enhancers, 0.29) and drives the cohn TATA motif-only and GC-content AUROCs to chance (0.371→0.500, 0.737→0.500). It collapses the cohn MCC (0.461→0.064), but random negatives of the same size do too (0.022), so we use AUROC: random reduction to the same 4.5% of cohn negatives lowers it 0.805→0.765, composition matching to 0.621, so sample loss is 22% of the drop in one matching draw (about 23% at the median of 20 draws). The drop persists in each design of Table II, and the gap grows with balance, then levels off. Matching also lowers the GC-balanced control, whose composition-only baseline falls as much when the model or the original test is held fixed: matching removes composition-

TABLE II

ROBUSTNESS OF THE DIAGNOSIS TO THE CONFOUNDS A SKEPTIC WOULD RAISE (k-MER LIGHTGBM AUROC ON `cohn` UNLESS STATED; 95% BOOTSTRAP CIS; RANDOM: MEAN OF 10 SIZE-MATCHED DRAWS IN k-MER ROWS; MATCHED: EQUALIZED, ONE DRAW). ALL SWEEP LEVELS, DATASETS AND DESIGNS ARE IN THE SM.

Concern	Test and result	Reading
Sample loss (4.5% kept)	Matched 0.621 vs. equal-size random 0.765 (all 0.805); gap +0.144 [0.102, 0.184]; sample loss 22% [12, 33] of the drop in this draw (about 23% at the median of 20 draws)	Mostly composition; size component
Test distortion; negative removal	Original test kept: 0.584 vs. 0.765 (+0.181 [0.166, 0.195]); original model on matched vs. random test: 0.595 vs. 0.803 (full test 0.805); positives matched, all negatives kept: 0.622 [0.585, 0.659] vs. 0.780	Drop persists
Matching threshold	Strictness α 0.02, 0.5, 0.9, 1 (93%, 18%, 5.8%, 4.5% kept): gap +0.001, +0.082, +0.160, +0.144; two-sided 1:1 caliper, 2 to 0.2 SD: +0.040 to +0.114	Grows with balance, then levels off
Selection bias	Enhancer-like cCREs: kept 34%, discarded 24%; cCRE-free test negatives: matched 0.646 vs. random 0.817	Enriched; drop persists
Artifact vs. biology	GC-balanced control: matched 0.674 [0.632, 0.719] vs. random 0.754; with the model or the test held fixed its composition-only baseline loses as much (0.076, 0.098 vs. k-mer 0.072, 0.051)	Not contamination-specific
Frozen features only	Fine-tuned, original→equalized (random) test negatives: HyenaDNA 0.796→0.581 [0.542, 0.618] (0.804), control 0.828→0.742 (26% of above-chance AUROC); DNABERT-2 0.848→0.720 [0.683, 0.752] (0.863)	Reliance survives; not contamination-specific
Homology: MCC	Train homolog: median 11% (max. 45%) of test sequences; purge vs. random removal flags 9 of 14 evaluable tasks, >0.05 MCC in 4 (post hoc cut; `nt_enhancers` −0.18, strength-matched −0.01; splice −0.06; `cohn` −0.01); cluster vs. random re-split, 3-seed means: coding-vs-intergenomic −0.09, splice −0.08	Real, uneven
Homology: labels	Labels kept: purge 14/14 evaluable (prevalence-adjusted MCC, post hoc)*; re-split 12/15*: `human_nontata_promoters` 0.64 vs. 0.87 random; NT enhancers solved under both	One relabel from homology
Homology: Route 1	ρ (11 tasks) +0.74→+0.55 (n.s.), pre-specified cluster re-split, seed 42 (random +0.57); 3-seed means: cluster +0.57, random +0.68, equal (+0.80) without NT-enhancer tasks; purge +0.83→+0.88 (n=10)	Random re-split lowers ρ too
Task covariates	Nested LOTO MAE, 9 pre-specified covariates: 0.143 vs. 0.140 discriminability (disc.) only (n.s.); disc.+length+GC SD+homology 0.233 (overfits); composition-only MCC (not a task property) 0.117 (n.s.)	GC-mean models tie; none beats disc.
Classifier (tuned LR / EBM)	14/15 labels each*; splice the only positional task (ΔMCC +0.68/+0.44); dose slope +1.95/+1.81 (LightGBM +1.99); real-task ρ +0.27/+0.47 (n.s.)	Changes where the weaker model fails
Third suite (GUE)	28 datasets, fixed thresholds: 12 failure-route, 8 solved, 8 no-single-route labels; only the 3 positional labels survive post hoc checks; 6 of 28 near-disjoint from GB/NT, none with a failure route (SM S8)	Scope narrowed

*Partly by construction: composition labels are task types; a model-free gate sets contamination vs. shared-motif; the purge re-scores only MCC and that gate.

associated signal, artifactual or biological. Because equalization drives the TATA and GC-content AUROCs to chance *partly by construction*, dinucleotide-shuffled negatives, which share the positives' composition, test whether the TATA skew is a by-product of the composition gap: the negative-class skew *vanishes* (`cohn` TATA 0.371→0.551, `nt_enhancers` 0.393→0.507; GC →0.500), the positives stay separable from their own shuffles by higher-order structure (k-mer AUROC 0.95/0.86), and for the control, whose TATA is a genuine positive motif, the TATA AUROC *rises* to 0.749 (Fig. 2).

Control-motif panel. Random PWMs matched in information content produce no class skew (motif-only AUROC 0.502±0.033 on `cohn`, 0.507±0.016 on `nt_enhancers`), but a column-scrambled TBP, which keeps its AT-rich column composition but not the ordered TATA pattern, stays skewed (0.376±0.017, 0.396±0.027): the skew is composition-driven, not a scan artifact. Real TATA on `nt_promoter_tata` skews to the positives (0.589) (SM S4).

TATA-flag cleaning. It removes 22.2% of `cohn` and 20.8% of `nt_enhancers` negatives (the excess counted over all sequences; none for the control, whose excess lies in its positives) and moves the TATA AUROC toward 0.5 (0.371→0.435; 0.393→0.485). The MCC drop (0.461→0.407; 0.514→0.459) is consistent with partial inflation (random removal of as many negatives lowers the `cohn` MCC by about 0.01).

Verdict by dataset. For `cohn` the pre-specified rule (equalized AUROC below every random draw, confirmed by overlap weighting and 1:1 caliper matching; SM S4) is met, with a size component, as it is for the control (without one); every model-free readout (motif scans, shuffles, control motifs, coordinates) is robust, but no model-level readout robustly separates `cohn` from the control beyond what a composition-only model shows (Section IV-F). For `nt_enhancers` the model-free signature is robust (coordinates only directionally), but the same rule is not met (36 equalized test negatives); on its leak-free test (post hoc), with the model or the original test held fixed, AUROC falls 0.15–0.20 more than with random negatives, and 1:1 caliper matching gives a `cohn`-like gap (0.594 vs. 0.729), but in each case a composition-only model loses as much. `human_enhancers_ensembl` rests on the TATA gate alone (Table I): post hoc, overlap weighting costs it a smaller share of above-chance AUROC than the control (LightGBM, LR; SM S7).

E. Frozen and fine-tuned FM probes

Frozen probes and fine-tuning test whether *pretrained* models exploit the shortcut; neither measures the Route-3 magnitude, which also includes genuine composition signal. A frozen probe's *test effect* is the AUROC change of the *same fixed head* when composition-biased test negatives are removed (retained GC-matched subset; nested paired bootstrap; negative = loss).

Frozen probes. With the [SEP] token, HyenaDNA loses significant enhancer AUROC (−0.048 [−0.053, −0.043] on `cohn` and −0.079 [−0.107, −0.056] on `nt_enhancers`), whereas the GC-balanced `drosophila` control is flat under the [SEP] token only, +0.005 [−0.003, +0.014] (with mean pooling, −0.039 against −0.058 on `cohn`). The effect is **head-invariant** (an LR head agrees). DNABERT-2's effects are near-identical (−0.047 and −0.072, mean-pool; all CIs exclude 0 for both poolings and heads). It was not run on

drosophila (a choice, not a context limit: only 2.3% of those sequences exceed 512 tokens; SM S1, S9); instead, removing as many cohn test negatives (1148) at random leaves it flat. The small test effect is a *floor*: on equalized negatives the same models lose 0.13–0.21 cohn AUROC, frozen or fine-tuned (SM S5).

End-to-end fine-tuning. The reliance survives: fully fine-tuned HyenaDNA-tiny and LoRA-tuned DNABERT-2 lose cohn AUROC when their test negatives are composition-equalized, but nothing when as many are removed at random (Table II). HyenaDNA loses 72% of its above-chance AUROC and DNABERT-2 37% (post hoc, same design: HyenaDNA not significantly different from the k-mer model's 66%, DNABERT-2 below its frozen mean-pool probe's 43%): LoRA reduces but does not remove the reliance. Without cCRE-overlapping test negatives (post hoc) the shares shrink to 56% and 24%. Retrained and tested on equalized rather than random negatives (pre-specified), DNABERT-2 reaches 0.703 vs. 0.833 on cohn; HyenaDNA's three cohn retrains collapsed to one-class fits (not evidence). On nt_enhancers (36 equalized test negatives) both lose AUROC (0.762→0.593; 0.820→0.680), but without cCRE-overlapping negatives the loss is no longer separable from random removal. The GC-balanced control (HyenaDNA) also loses 0.086 AUROC, against 0.169 on nt_enhancers (CIs overlap), failing the pre-specified rule that it lose clearly less than both enhancer sets; fine-tuning, like matching, shows reliance on composition-associated signal, not that it is an artifact.

F. Robustness

Homology. Homology can inflate only the model-scored labels, solved and positional (Table II): human_nontata_promoters passes the purge only on the prevalence-adjusted MCC (purged test re-weighted to the original class proportions; post hoc) and relabels under cluster-aware re-splits; the NT enhancer tasks become solved under random re-splits too (re-pooling, not homology). On homology-clean data, Route 2 persists (splice ΔMCC +0.62 vs. +0.60 original, equal-size training subsamples), as do, on cohn, TATA-flag cleaning (TATA AUROC 0.435 vs. 0.373 for random removal), equalization (k-mer AUROC 0.55–0.63 over 20 matching draws, all below 0.737 for random removal), the control-motif panel, the TSS depletion and the FM shortcut (frozen; fine-tuned HyenaDNA); on homology-clean nt_enhancers the equalized splits keep 3 test negatives (not evaluable) and fine-tuned HyenaDNA's GC-matched test effect vanishes (SM S6).

Classifier. Each label change under the MCC-tuned LR or the EBM is a solved task the weaker model fails: human_ensembl_regulatory under LR and human_nontata_promoters under the EBM. No model-level Route-3 readout is contamination-specific: under overlap weighting cohn loses more than the control under every model, but for LightGBM its composition-only baseline shows the same contrast, and, retrained on equalized vs. random negatives, the cohn-minus-control difference is significant only for the MCC-tuned LR, not when the same LR is tuned by AUROC (SM S7). The k-mer representation was not varied (no gapped-k-mer SVM).

V. Discussion

Benchmark design, not dataset corruption. Composition alone supplies most of the separability of two hard enhancer benchmarks (AUROC 0.78/0.82 vs. k-mer 0.80/0.85), a TATA/AT-rich signal, not core-promoter leakage (only directionally shown for nt_enhancers), that our classifier and two FMs exploit to different degrees. Most of the composition gap is the positives' GC-richness, not the negative sampling, and no model-level design shows that the removed composition signal is an artifact rather than biology. Neither source is a data error; the critique is one of *benchmark design*: a composition axis does not test regulatory grammar.

Recommendations. (i) Equalize negative-set composition to the positives (or use dinucleotide-preserving shuffles, keeping each positive and its shuffle in the same split) and benchmark the cleaned split against size-matched random negatives; since equalization also removes genuine composition signal, report both splits. (ii) Report a composition-only baseline. (iii) Ship a route label and a homology-redundancy audit in the benchmark card. (iv) Evaluate a position-aware baseline before concluding a representation has failed. Our released CLI produces such a route-labeled report card (on its development tasks it reproduces 11 of the 12 Table I labels it can emit).

Limitations. (i) Fine-tuning covers two modest FMs, one untuned recipe each (HyenaDNA below leaderboard accuracy: cohn 0.716 vs. 0.742 [4]); the GC-balanced control covers HyenaDNA only, and the equalized cohn test negatives overlap cCREs more often than a size-matched random draw (50% vs. 32%). Full-parameter DNABERT-2 and NT-v2-500M fine-tuning is packaged but was not run (SM S9). (ii) Matching removes composition-associated signal indiscriminately (also from the control) and incompletely (composition-only AUROC 0.581 on our released equalized cohn split). (iii) Homology-clean fine-tuning covers HyenaDNA only, and the dose-response is not re-run (no natural homology). (iv) Data: nt_enhancers is mostly RankGAN-synthetic (Section IV-D); cohn has no chr9 negatives against 600 chr9 positives, an unexplained skew and a potential label shortcut. (v) Scope: on GUE, which its authors filtered for difficulty, only the 3 positional labels survive our post hoc checks, three of the six applicable pre-stated predictions failed, and only 6 of the 28 datasets are near-disjoint from GB/NT (SM S8).

Artifact and reproducibility. Zenodo holds the frozen-FM embedding cache (doi: 10.5281/zenodo.20494982). Code, the SM, the diagnose.py CLI and composition-equalized splits for cohn, nt_enhancers (keeping its 100 test positives that duplicate training sequences) and the drosophila control are at https://github.com/rikhinkavuru/gb_kmer_benchmark.

VI. Conclusion

We presented a CPU-only diagnostic framework that route-labels the tasks of two genomic benchmark suites into three routes to failure — shared-motif, positional and contamination

— each established by intervention. Two enhancer benchmarks carry a composition gap between GC-rich positives and AT-richer negatives, a *benchmark-design* confound, not dataset corruption and, for `cohn`, not core-promoter leakage. Two architecturally unrelated FMs, frozen or fine-tuned, lose significant enhancer AUROC when composition-biased negatives are removed, but not when as many are removed at random; since a GC-balanced control also loses AUROC (HyenaDNA; fine-tuned, or frozen mean-pooled), this shows reliance on composition-associated signal, not that the signal is an artifact.

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